



FACULTY OF BUSINESS AND IT

MBAI 5600G - Applied Integrative Analytics Capstone Project

Milestones 8

Due: August 11, 2026

Milestone	Description	Due Date
1	Capstone project selection/proposal	May 19
2	Literature review and data source identification	May 26
3	Data collection & Initial EDA	June 9
4	Data Preprocessing & Baseline Modeling	June 23
5	Advanced Modeling & Optimization	July 7
6	Validation & Business Impact Assessment	July 21
7	Capstone project oral presentation	July 27 & August 10
8	Capstone project report	August 11

Important Notes for the Final Capstone Project Report

Please ensure your final report addresses the following requirements:

- **Describe all tables and figures.** Every table and figure must be referenced and discussed in the main text. Explain what it shows, highlight the key findings, and discuss its significance in relation to your project objectives.
- **Include a workflow or methodology diagram.** Provide a clear flowchart or system workflow illustrating your proposed methodology. This helps improve readability and enables readers to understand the overall process and system architecture more easily.
- **Avoid unnecessary repetition.** Present each concept clearly and concisely. Refrain from repeating the same explanations, discussions, or findings in multiple sections of the report unless necessary for clarity.
- **Define all evaluation metrics and computed quantities.** For every performance metric (e.g., Accuracy, Precision, Recall, F1-score, AUC-ROC, RMSE, MAE, etc.) and any other computed quantity used in your analysis, provide a brief description, the corresponding mathematical formula, and explain its purpose. All equations should be numbered and referenced appropriately within the text.
- **Perform statistical significance testing.** Where model comparisons are presented, include appropriate statistical significance testing to determine whether observed performance differences are statistically meaningful. Clearly describe the test used, the rationale for selecting it, and interpret the results in the context of your project.

Milestone 8: Capstone Project Final Report

As the final component of your capstone project, you are required to submit a comprehensive **final report** along with a complete **package containing your code and the dataset** used. This milestone represents the culmination of your project journey—from initial concept to full implementation—and serves as a demonstration of your ability to apply advanced machine learning and AI techniques.

Your report should clearly detail the project's objectives, the methodologies used, key findings, and the overall impact of your work. The accompanying code and dataset package should be well-organized and reproducible, enabling others to replicate your results and potentially extend your work, thereby contributing to the broader machine learning and AI community.

I. Final Report

To prepare your final report, use the provided template file (**Capstone Project Report Template.docx**) along with the reports you've submitted for each milestone throughout the semester. As you complete your final document, follow the structure outlined below.

Title: Provide a concise and descriptive title that captures the essence of the project.

Student Names: [Full Names]

Abstract: The abstract should briefly state the purpose of the research, the principal results, and major conclusions. An abstract is often presented separately from the article, so it must be able to stand alone. For this reason, References should be avoided, but if essential, then cite the text. Also, non-standard, or uncommon abbreviations should be avoided, but if essential, they must be defined at their first mention in the abstract itself.

Keywords: Immediately after the abstract, provide a maximum of 6 keywords, using American spelling and avoiding general and plural terms and multiple concepts (avoid, for example, 'and', 'of'). Be sparing with abbreviations: only abbreviations firmly established in the field may be eligible. These keywords will be used for indexing purposes.

1. **Introduction:** Provide a brief introduction to the project by describing the business context, the motivation for the study, and the overall objective. Explain the importance of the problem and provide sufficient background to help readers understand the purpose and scope of the project.
2. **Problem Statement:** Clearly define the business problem or opportunity addressed by the project. Explain why it is important, identify the key challenges, and state the project objectives and expected outcomes. Ensure the problem statement is specific, measurable, and aligned with the proposed AI or analytics solution
3. **Background & Literature Review:** A review of existing literature, research papers, reports, and relevant sources related to your project.
4. **Data Description:** Describe the dataset used in the analysis, including its source, size, variables, and any notable characteristics or limitations.
 - 4.1 **Methodology:** Explain the approach and techniques used in the data analysis, such as data collection methods, data cleaning and preprocessing techniques, and the analytical tools or software employed.
 - 4.2 **Exploratory Data Analysis (EDA):** Present an overview of the EDA performed on the dataset, including visualizations, summary statistics, and key insights gained from the exploration.

4.3 Limitations: Discuss any limitations or challenges encountered during the data analytics process, such as data quality issues, data gaps, or constraints that may have affected the analysis and interpretations.

5. **AI/ML Models:** Provide a detailed description of AI/ML models utilized in your project. Furthermore, discuss any modifications made, if applicable, to these models and provide an overview of the parameter settings employed in your experiments. This section should describe baseline model/s, model training, hyperparameter tuning, model validation, model comparison, and evaluation metrics.

6. **Experimental Results and Analysis:** Here, you should provide results obtained from your prediction models. It should summarize the key insights derived from the data analysis and provide actionable recommendations based on those insights. These recommendations should aim to address specific challenges or opportunities identified in the analysis.

It is crucial to recognize the significance of the experimental results section in any research or project report. This section serves as a vital component by providing empirical evidence and conducting a thorough analysis of your experiments or investigations. Through this section, readers can evaluate the validity and effectiveness of your work, gaining valuable insights into the outcomes, findings, and data acquired from your experiments, simulations, or tests.

7. **Business Impact:** Discuss how the proposed solution creates value for the organization by explaining its expected impact on business performance. Wherever possible, quantify the potential business benefits, such as cost savings, revenue growth, improved operational efficiency, enhanced customer experience, or risk reduction. In addition, explain how the findings can support managerial decision-making by providing actionable insights. Finally, discuss the strategic implications of implementing the proposed solution, including its potential long-term benefits, alignment with organizational objectives, and contribution to maintaining a competitive advantage.

8. **Ethical and Responsible AI:** Discuss the ethical and responsible AI considerations relevant to your project by addressing potential sources of bias in the data, model, or decision-making process and explaining how these biases can be identified and mitigated. Evaluate the fairness of the proposed solution, including its potential impact on different users or stakeholder groups. Explain how privacy is protected throughout the data collection, processing, and deployment stages, with appropriate consideration of data security and regulatory compliance. Finally, discuss the governance mechanisms that support the responsible development and use of the AI solution, including accountability, transparency, monitoring, and oversight throughout its lifecycle.

9. **Limitations and Future Work:** Discuss the key limitations of the project and explain how they may have affected the results, performance, or applicability of the proposed solution. Reflect on any constraints related to the data, methodology, model design, computational resources, or evaluation process. Based on these limitations, identify potential directions for future work, including opportunities to improve the model, incorporate additional or higher-quality data, enhance scalability and robustness, explore alternative methods, or extend the solution to new applications or business contexts. Conclude by explaining how these future enhancements could strengthen the overall effectiveness and impact of the proposed solution.

10. **Conclusions:** Summarize the main findings and outcomes of the data analytics project, emphasizing the value and implications for the business. Furthermore, outline the proposed next steps for the

project, including potential further analysis, data collection, or refinement of the analytical approach. This section should provide a roadmap for moving forward with the project.

11. **References (IEEE Style):** Students must use proper IEEE citation formatting consistently throughout the document.

[1] J. Smith and A. Kumar, "Explainable ensemble learning for customer churn prediction," *IEEE Access*, vol. 13, pp. 11234–11249, 2025.

[2] R. Chen et al., "Comparative analysis of deep learning and traditional machine learning methods for financial forecasting," *Expert Systems with Applications*, vol. 245, 2026.

Appendices (if applicable): Include any supplementary material, such as additional data, charts, graphs, or code snippets, that are relevant to understanding your report.

II. Code/Dataset Package

Please ensure that your final project code/data is submitted in a zipped folder, comprising the following essential files and datasets:

1. **Source code:** Include bug-free and well-documented source code, allowing others to understand and replicate your project.
2. **Readme file:** Include clear, step-by-step instructions for installing and running your project, detailing any required dependencies or specific configurations. Refer to the **README File Instruction.docx** for guidance.
3. **Dataset used:** Include all raw data utilized in your project. If the data is large, it is advisable to submit the datasets to an appropriate public data repository and provide the necessary links or instructions to access them.

Submission Guideline

Please submit your final project report in PDF or DOCX format, along with your code/data package as a ZIP file, **separately** by **August 11th at 11:59 PM**.